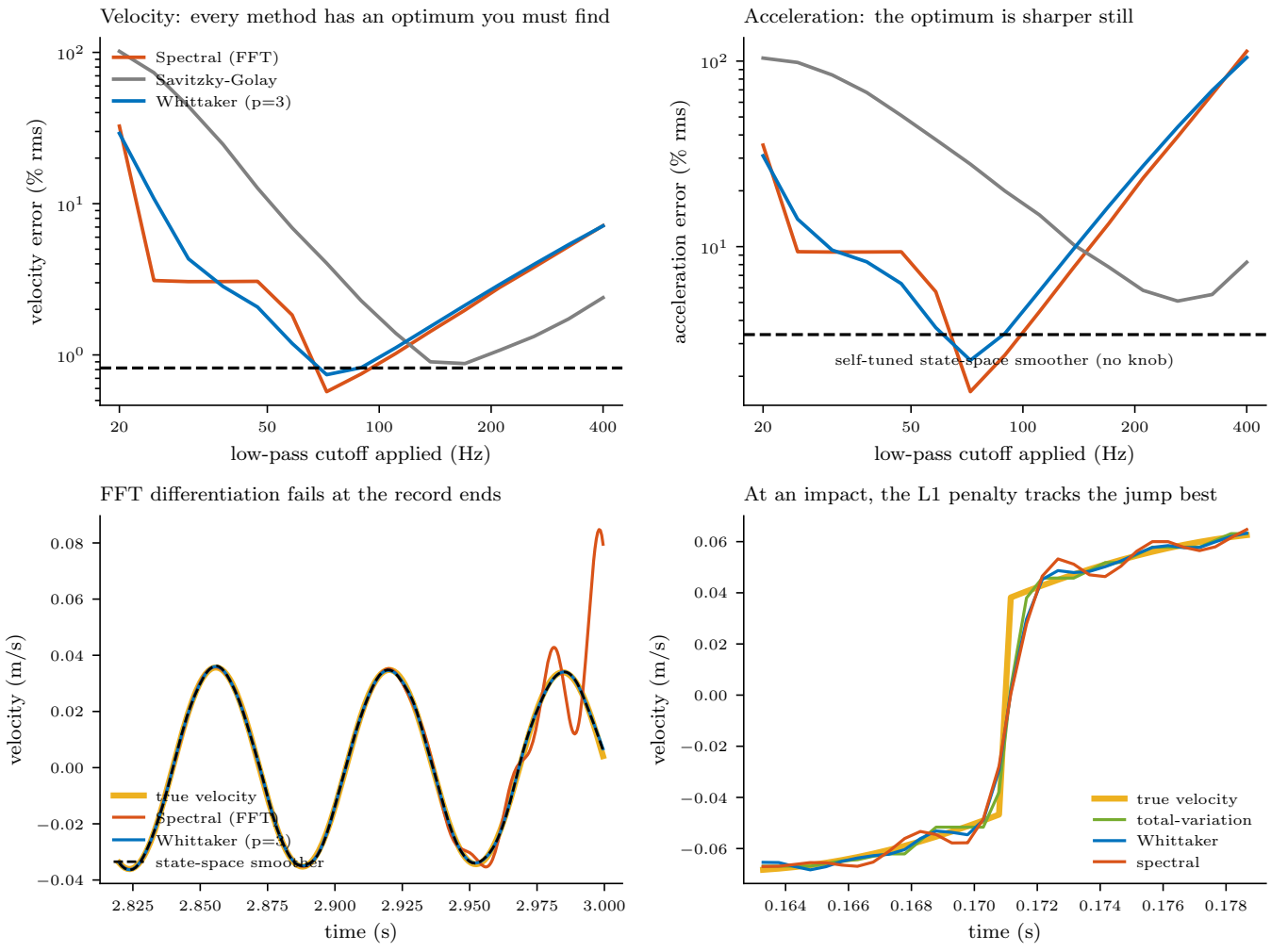

Velocity: every method has an optimum you must find
Spectral (FFT)
Savitzky-Golay
Whittaker (p=3)
velocity error (% rms)
low-pass cutoff applied (Hz)
Acceleration: the optimum is sharper still
acceleration error (% rms)
self-tuned state-space smoother (no knob)
low-pass cutoff applied (Hz)
FFT differentiation fails at the record ends
true velocity
Spectral (FFT)
Whittaker (p=3)
state-space smoother
velocity (m/s)
time (s)
At an impact, the L1 penalty tracks the jump best
true velocity
total-variation
Whittaker
spectral
velocity (m/s)
time (s)